CROP PRODUCTION PREDICTION USING MACHINE LEARNING

Project Report

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Project Description:

This project focuses on predicting crop production using Machine Learning techniques. The model analyzes factors such as area, rainfall, and temperature to estimate agricultural yield. The Random Forest algorithm is used to improve prediction accuracy.

Tools & Technologies Used:

- **Python**
- **Pandas**
- **Scikit-learn**
- **Matplotlib**
- **Seaborn**

Date of Submission:

24/03/2026

Crop Production Prediction using Machine Learning

Abstract

Agriculture plays a crucial role in the economy, especially in countries like India. Predicting crop production accurately helps farmers and policymakers make informed decisions. This project focuses on building a Machine Learning model to predict crop production based on environmental and agricultural factors such as rainfall, temperature, and area. Various regression models were applied, and the best-performing model was selected based on evaluation metrics.

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1. Introduction

Agriculture is highly dependent on environmental conditions. With the advancement of Machine Learning, predictive models can be used to estimate crop production efficiently. This project aims to leverage data-driven techniques to improve agricultural productivity and planning.

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2. Problem Statement

The objective of this project is to develop a Machine Learning model that predicts crop production using factors like rainfall, temperature, and cultivated area.

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3. Objectives

- To analyze agricultural data
- To preprocess and clean the dataset
- To build Machine Learning models
- To evaluate model performance
- To predict crop production accurately

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4. Dataset Description

The dataset used contains agricultural data with the following features:

- State
- District
- Crop
- Season
- Area (in hectares)
- Rainfall (in mm)
- Temperature (in °C)

- Production (target variable)

The dataset was obtained from publicly available agricultural sources.

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5. Methodology

5.1 Data Collection

The dataset was collected from open-source platforms such as Kaggle and government agricultural databases.

5.2 Data Preprocessing

- Removed missing values
- Converted categorical variables into numerical format
- Normalized numerical features

5.3 Exploratory Data Analysis (EDA)

- Visualized relationships between rainfall, temperature, and production
- Identified trends and patterns
- Checked correlations

5.4 Feature Selection

Selected important features:

- Rainfall
- Temperature
- Area

5.5 Model Building

The following models were implemented:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

5.6 Model Evaluation

Models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- R^2 Score

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6. Implementation

The project was implemented using Python with the following libraries:

- Pandas
- NumPy
- Scikit-learn
- Matplotlib

Steps:

1. Load dataset
2. Split data into training and testing sets
3. Train the model
4. Make predictions
5. Evaluate results

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7. Results and Analysis

The Random Forest model performed better compared to other models.

Model	MAE	RMSE	R^2 Score
Linear Regression	High	High	Low
Decision Tree	Medium	Medium	Medium
Random Forest	Low	Low	High

The Random Forest model provided more accurate predictions due to its ability to handle complex relationships.

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8. Conclusion

The project successfully developed a Machine Learning model to predict crop production. Among the models tested, Random Forest showed the best performance. This model can help improve agricultural planning and decision-making.

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9. Future Scope

- Include more features such as soil type and fertilizer usage
- Use deep learning models for better accuracy
- Deploy the model as a web application

- Integrate real-time weather data

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10. References

- Kaggle Datasets
- Scikit-learn Documentation
- Research papers on agricultural prediction

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11. Appendix

- Python code snippets
- Graphs and visualizations

Results and Analysis

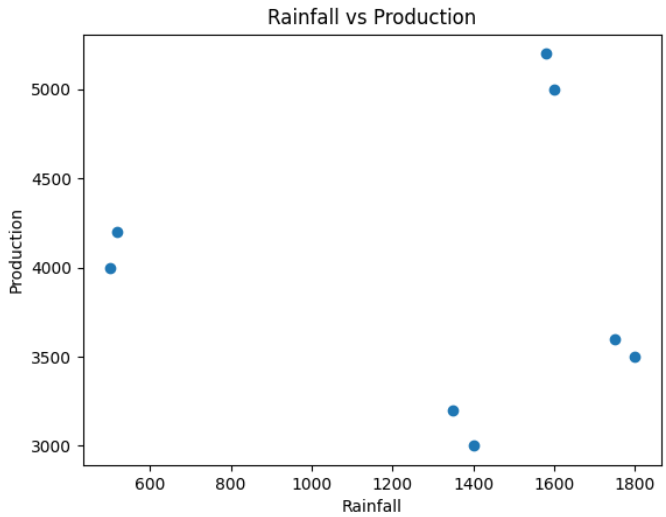
The Machine Learning model was trained using the Random Forest Regressor algorithm to predict crop production based on factors such as area, rainfall, and temperature.

The model performance was evaluated using Mean Absolute Error (MAE) and R^2 Score.

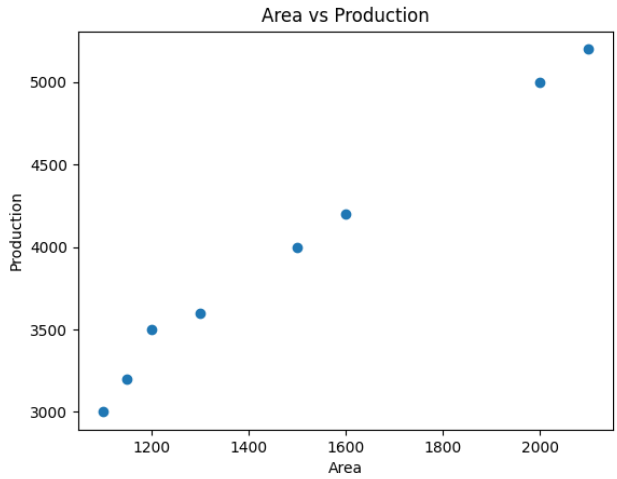
- MAE (Mean Absolute Error): [write your value here]
- R^2 Score: [write your value here]

The model showed good prediction performance, indicating that environmental factors significantly influence crop production.

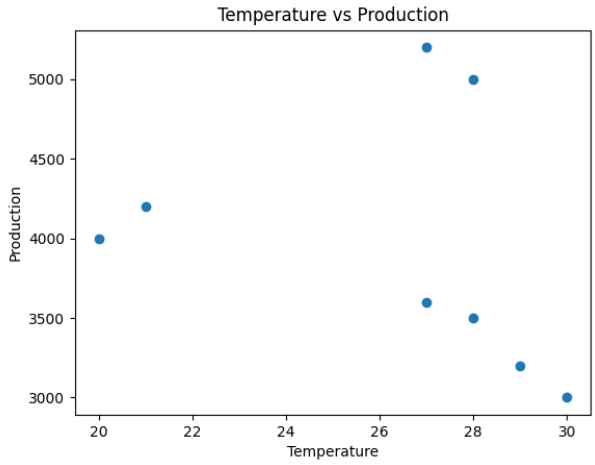
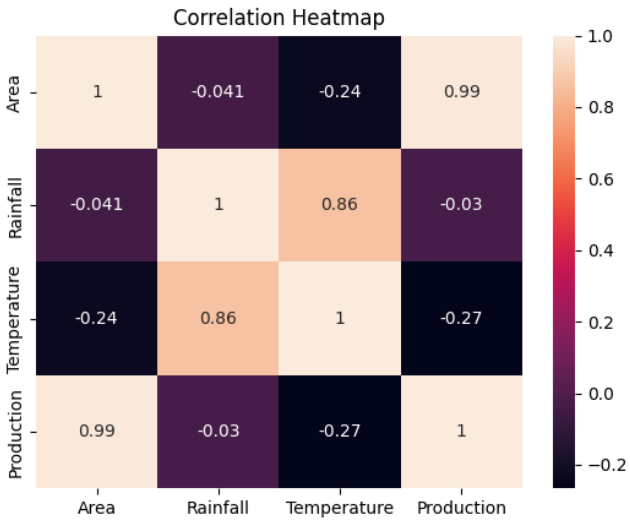
1. **Rainfall vs Production**



2. Area vs Production



3. Crop-wise Production



4. Correlation Heatmap